

Does urban overheating affect neighborhood attractiveness? Evidence from the French housing market Revised Version 1

Pierre VINCENT

Univ Lyon, UJM Saint-Etienne, GATE UMR 5824, F-42023 Saint-Etienne, France

February 2025

Abstract

This article examines the relationship between urban overheating and housing prices, positioning urban overheating as an environmental dis-amenity with direct implications for the quality of life and health of urban residents. By reducing the attractiveness of affected neighborhoods, urban overheating is expected to negatively affect housing prices. A key contribution of this study is the use of the original Thermocity dataset, which provides high-resolution nighttime land surface temperature data derived from satellite observations over a set of four French urban areas. Housing data are sourced from an exhaustive dataset of housing transactions for the period 2013-2019, which provides detailed information on housing prices and characteristics. To address the potential endogeneity of local urban overheating, an instrumental variable approach is employed, leveraging wind speed as an instrument for land surface temperature. Results from the 2SLS estimation reveal a significant negative effect of urban overheating on housing prices: a 1°C increase in urban overheating is associated with a 4.2% decrease in housing prices. Moreover, the analysis highlights the variation in the effect across urban units and over time, with a stronger effect observed in the latter part of the study period. The robustness of these results is confirmed using alternative measures of urban overheating from other data sources.

JEL codes - C36, Q51, R31

Keywords - housing prices, hedonic models, urban overheating, amenities, climate change, land surface temperature

Contact:

Univ Lyon, UJM Saint-Etienne, GATE UMR 5824, F-42023 Saint-Etienne, France

Permanent address: 81 rue Michelet, 42000 Saint-Etienne

Email: vincent@gate.cnrs.fr

1 Introduction

In urban areas, the combination of increasing urbanization and global warming is leading to an increase in the incidence of urban overheating, which is characterized by air temperatures that are significantly higher than those in nearby rural areas. With nearly 81.5% of the French population living in urban areas, urban overheating concerns a large proportion of the French population. During the summer months, many metropolitan centers, such as the center of Paris, suffer from high temperatures. This localized rise in temperature affects not only thermal comfort but also public health, energy consumption, air quality, and the economic attractiveness of urban neighborhoods.

Urban overheating results from specific meteorological conditions that are becoming more frequent due to climate change. However, its primary drivers are the physical characteristics of urban environments, such as the building density, the thermal properties of building materials and low vegetation cover. In addition, anthropogenic activities, including transportation and the use of air conditioning, exacerbate the intensity of urban overheating. As a result, urban overheating is a highly localized phenomenon that can be observed at the neighborhood level. Temperature variations can be striking, sometimes varying significantly from one street to another, depending on the differences in physical features of the urban landscape. (Nazarian et al., 2022)

Questions arise regarding the capacity of households to cope with urban overheating, particularly in relation with two types of adaptation: housing investment and location choice. Housing investments may include energy-efficient retrofits or air conditioning systems, which, although effective, are costly and primarily address indoor environments. Alternatively, households can adapt by choosing to reside in areas less susceptible to urban overheating. This shift in location preferences could potentially lead to a change in the urban spatial equilibrium, with cooler neighborhoods becoming more appealing due to their more comfortable living conditions compared to hotter areas. The impact of urban overheating on household location decisions could be substantial, potentially influencing housing prices at the neighborhood level and reshaping the spatial distribution of households. Wealthier households often have the ability to secure access to cooler neighborhoods, while poorer households may find themselves confined to the hotter parts of the city. This study aims to address this knowledge gap by estimating the impact of urban overheating on housing prices, providing valuable insights into how urban overheating reshapes preferences for neighborhoods.

In addition to providing one of the first evaluations of economic literature on urban overheating, this article contributes to the recent literature on climate change within regional and urban economics. The hedonic pricing method has been widely used in the literature to address the value of certain amenities, such as climate. Climate valuation through hedonic price models emphasizes that households prefer cool summers and mild winters (Maddison and Bigano, 2003; Rehdanz, 2006; Rehdanz and Maddison, 2009). While these results include a concept of overheating, the existing work focuses on inter-city analyses, whereas urban overheating, as I intend to address it, is an intra-city phenomenon. The examination of this literature also reveals that the climate variables used are large-scale measures, often involving average temperature measured over a month at the scale of an urban area or a county. For instance, Grout et al. (2016) employed average July and January temperatures at the urban areas level to analyze the impact of climate on urban sprawl.

In this study, urban overheating is taken as an environmental dis-amenity that has an implicit price for households, manifesting in housing prices. A major novelty of this paper is the use of a highly localized measure of urban overheating from earth science: data from the Thermocity project (Michel et al., 2022). Thermocity data consist in nighttime satellite observations of Land Surface Temperature (LST, hereafter) at the resolution of 70 meters, covering five French urban units.¹ Designed specifically for the study of urban heat islands within these French urban units, the Thermocity LST dataset offers unprecedented granularity, which allows for the identification of local variations in the urban climate and the inclusion of fine-scale fixed effects that control for unobserved variables impacting neighborhood attractiveness. To the best of my knowledge, the use of such microclimatic data within the field of urban economics represents a significant novelty and constitutes a substantial contribution of this study.

The main challenge confronting this analysis is to achieve an unbiased estimation of the impact of urban overheating on housing prices. Indeed, the OLS estimation of urban overheating effects via a hedonic pricing model might be biased due to endogeneity, emerging from various sources. To address endogeneity stemming from the correlation

¹Due to housing prices data availability, only four urban units are included in the following analysis.

of urban overheating with numerous unobserved factors and potential measurement errors, an instrumental variable approach is adopted, with the wind speed as an instrument, aggregated at a reference scale of $250\text{m} \times 250\text{m}$. Actually, wind speed plays a crucial role in mitigating urban overheating by contributing to the cooling of the urban environment.

The results of the 2SLS estimation indicate that urban overheating has a negative impact on housing prices. Results with the preferred specification suggests that an increase of 1 degree in the land surface temperature is associated with a decrease by 4.2% in housing prices. Estimates by urban units show that urban overheating has different impacts depending on the city. Estimated coefficients range from -0.021 for Toulouse, to -0.102 for Marseille. Estimations by subperiods show that this effect has increased in the second part of the studied period. Robustness checks were conducted to address concerns related to unobserved factors and the validity of the Thermocity measure of urban overheating. Regressions using the alternative urban overheating measures from UrbanSIS project (Lauwaet et al., 2024) tend to confirm the main analysis results.

The organization of this article is as follows. Section 2 provides an overview of the related literature. Section 3 defines urban overheating and provides contextual background. Section 4 presents data used for the analysis. Section 5 exposes the empirical strategy. Section 6 shows the main results based on 2SLS estimations. Section 7 provides robustness checks. Finally, section 8 concludes.

2 Related literature

The study of the determinants of neighborhood attractiveness constitutes a fundamental research topic in regional and urban economics. Among the numerous factors examined, those related to the environment hold particular significance. To quantify the effect of an amenity on household location choices, it is common to use hedonic pricing models, which aim to reveal and quantify household preferences, based on their impact on housing prices. Recently, research using hedonic pricing models to value an amenity combines it with precise identification methods, using exogenous sources of variation in amenity level. Hennighausen et al. (2023) use an exogenous variation in insurance legislation to reveal households preferences with respect to flood risk. Choi et al. (2023) use a gas leak to address the effect of air quality on housing prices. I contribute to this literature by also using a hedonic pricing model for housing to quantify the effect of an environmental dis-amenity, namely urban overheating, on housing prices.

While, to the best of my knowledge, there is no literature dealing with urban overheating effect on housing prices, several bodies of the literature that usually use hedonic pricing models on related issues namely climate valuation and green amenity valuation. Hedonic pricing models are widely used in the climate valuation literature, as housing is a long-term investment that raises concerns about climate risks. Maddison and Bigano (2003), Rehdanz (2006), and Rehdanz and Maddison (2009) investigate climate valuation by households in three European countries with distinctly different climates: Italy, Great Britain, and Germany, respectively. Hoehn et al. (1987) and Koirala and Bohara (2014) both study the effect of climate on housing prices across the United States based on American Census data. Butsic et al. (2011) focus on the specific case of housing price evolution in American ski resorts in the perspective of climate change. Ma and Yildirim (2023) focus on the impact of local heat shock on housing price and climate change belief in the USA between 2000 and 2021. Kang et al. (2024) study the impact of the frequency of extreme heat events on housing prices in Chinese cities from between 2009 and 2019.

Other methods have been used in the literature to quantify the value of climate. Cheung et al. (2018) use an artificial network approach to examine the effect of temperatures on residential skyscraper prices in Hong Kong, known for their high density and vulnerability to heat waves. Cvijanovic and de Minne (2023) examine the effect of exposure to extreme meteorological events on commercial real estate investment performance in the USA between 1980 and 2020. Sinha et al. (2018) use a discrete choice approach based on American census data to calculate the marginal willingness to pay of households for climate. Unlike the hedonic pricing approach, which only requires data on housing, discrete choice models require household location choice data that are less often available than housing data. Even if methodology and climate measures differ from an article to another, those articles reach similar conclusions. Extreme temperatures are perceived negatively by households. Conversely, households perceive higher winter temperatures positively. In this sense, these articles are slightly linked to the notion of overheating. They also demonstrate the heterogeneity of household preferences: contrary to households living in warm areas, households in colder areas do

not consider an increase in summer temperatures as deleterious.

The topic of urban overheating can also be related to another body of literature: the valuation of environmental amenities, such as green spaces or open spaces, which often have a cooling effect on their urban surroundings. This literature also uses hedonic housing price models to determine the marginal willingness to pay of households for environmental amenities. For instance, Nilsson (2014) focuses on the valuation of open spaces in Sweden and highlights the importance of these amenities for households. Additionally, a portion of this literature focuses on the particular case of urban tree valuation, also studied through hedonic housing price models. Some studies use exogenous variation of tree cover in order to identify a causal effect of trees on housing prices. This is the case of Bouwknecht and Rouwendal (2023) and Han et al. (2024). Especially, the latter base their empirical strategy on an exogenous insect invasion in Toronto that killed ash trees. Other geographical areas have been considered in articles aiming to quantify the value of trees for households. Sander and Haight (2012) focus on the Dakota County in Minnesota; Li et al. (2022) on the Chinese real estate market; and Luttik (2000) on the Netherlands. However, green amenities offer various benefits to households, so that households value these amenities for different reasons, and not specifically due to their cooling effect, leading to identification issues when interested in a specific dimension of green amenities.

Although urban overheating is a great concern for other scientific fields, to the best of my knowledge, there is no research work estimating the value of urban overheating to households. Grislain-Letrémy et al. (2024) conducts extensive descriptive work by examining the exposure of different household groups to urban overheating in 10 major French cities. They find that household exposure varies by income across cities. For example, their results show that in Lyon, exposure to overheating decreases with income, whereas in Paris, both the very rich and the very poor households are exposed. Despite the lack of specific work about urban overheating effect on housing prices, it is worth noticing that the impact of climate change on location of economic activities and urban configurations is a growing field of research in spatial and urban economics. Grout et al. (2016) show that cities with warmer climates are associated with greater urban sprawl in France. There is a more recent literature using urban quantitative spatial models to quantify the cost of climate change in terms of production, according to regions of a country, or regions of the world (Desmet and Rossi-Hansberg, 2012; Bilal and Rossi-Hansberg, 2023).

To conclude this section, it is worth noticing that all the previously mentioned works use relatively large-scale climate measures, based on average air temperatures over a period at the regional or city level. For example, Grout et al. (2016) use average July and January temperatures at the level of urban areas. Nevertheless, the trend seems to be towards more precise and temporally variable measurements of the climate, such as in Cvijanovic and de Minne (2023), where authors combine daily temperature and precipitation at a 4km resolution from PRISM Climate Group to compute an extreme temperature exposure index. However, using such data is not relevant for the study of urban overheating, which requires a within-city measurements at the neighborhood level. One important contribution of this article is the use of very fine microclimatic data to quantify urban overheating.

3 On urban overheating

In France, urban overheating is increasingly recognized as a serious public concern. Since more than 80% of the French population lives in urban areas, the majority of the population might be affected by the problem. Nearly 30% of French people live in one of the 10 largest urban units in France, which are also the most affected by urban overheating. In Occitania region, according to INSEE, half of the residents are frequently exposed to situations of extreme heat.²

Providing a precise definition of urban overheating is a delicate task. For this study, I adopt the definition proposed by Nazarian et al. (2022), who define urban overheating as *the exceedance of locally defined thermal thresholds that correspond to negative impacts on people (e.g., health, comfort, and productivity) and associated urban systems*. Operationally, measuring urban overheating involves comparing the air temperature of the urban zone with the temperature of the surrounding rural area. Nighttime measurements, specifically called urban heat islands, are particularly preferred to daytime temperatures, as they reveal the extent to which the urban environment absorbs and is unable to dissipate heat naturally compared to the rural environment.

²<https://www.insee.fr/fr/statistiques/4299803>

Numerous factors contribute to the formation of urban overheating, which can be categorized into two groups: controllable and uncontrollable factors (Rizan et al., 2008). Uncontrollable factors pertain to meteorological conditions conducive to urban overheating, such as cloud cover, anticyclonic conditions, or wind speed. These uncontrollable factors wield considerable influence over urban overheating dynamics. Additionally, the frequency of meteorological conditions that contribute to urban overheating is increasing due to accelerating climate change. Controllable factors, on the other hand, are directly influenced by human actions. Firstly, urban morphology elements such as density, form, and composition of the urban environment play a crucial role. The materials utilized in construction directly impact the absorption of solar radiation by surfaces, consequently affecting surface and air temperature, and perceived temperature. For instance, dark materials like asphalt absorb substantial solar radiation, leading to significant heat absorption during hot periods, whereas lighter surfaces reflect more radiation, resulting in cooler surface temperatures. Impermeable surfaces and the lack of green spaces hinder cooling through evapotranspiration. Furthermore, the design of buildings influences urban overheating by restricting wind circulation in streets and limiting available shading throughout the day. Secondly, human activities contribute to urban overheating, with air conditioning and transportation being notable sources of heat emission that can elevate local temperatures. Analyzing the combined provision of these factors neighborhood by neighborhood provides an indication of how conducive an area may be to urban overheating formation.

Urban overheating poses numerous challenges for exposed residents. Consequences of urban overheating encompass four themes: health, productivity, air quality, and energy consumption. The most apparent concern is the public health issue induced by high temperatures. Extreme heat clearly has a direct impact on people’s health. Nazarian et al. (2022) highlights the fact that the cooling mechanism of the body puts pressure on the cardiac system, a pressure that can be burdensome for vulnerable populations such as the elderly and young children. Sweating, another body’s cooling mechanism, can also lead to dehydration. *Santé Publique France* (French Health instance) estimates that each year, between June 1st and September 15th, 33,000 people die due to heat.³ Beyond the direct health risks, high temperatures have a real impact on individuals’ quality of life and productivity. In their reviews of the literature about the consequences of urban heat island and urban overheating, Santamouris (2020) highlights the adverse effects of overheating on air quality and energy consumption, since elevated temperatures are associated with increased cooling demand, leading to a rise in electricity consumption. Moreover, high heat promotes the formation of toxic molecules such as tropospheric ozone. Induced electricity production also results in higher pollutant emissions from power plants, exacerbating air quality concerns. Beyond these direct effects, widely studied in earth sciences, urban overheating might also affect the urban real estate market, as it deprecates the neighborhood attractiveness.

4 Data

This section presents the data used in this study and explains the construction of the housing database. The final database contains observations of dwellings sold between 2013 and 2019 in the urban units of Marseille, Montpellier, Paris, and Toulouse. Subsection 4.1 focuses on the urban overheating data, while Subsection 4.2 describes the housing transaction data. Finally, Subsection 4.3 outlines the steps undertaken to compile and structure the database.

4.1 Urban overheating data

As mentioned in the previous section, defining urban overheating and finding precise data to measure it is challenging. The primary requirement for a measure of urban overheating is the spatial resolution. Unlike the large scale climate measurements used in the climate valuation literature, urban overheating measurements need to be fine-scaled, ideally at the neighborhood level. Urban overheating is generally characterized by the temperature differential between specific urban locations and the surrounding rural areas. However, systematic recordings of air temperatures at neighborhood level does not exist in France.

Land Surface Temperature (LST), derived from satellite observations at specific time points, is a widely used metric in earth sciences to analyze urban overheating (Nazarian et al., 2022; Zhou et al., 2019). LST is influenced by surface properties such as thermodynamic characteristics, moisture levels, and exposure to solar radiation (Naserikia et al., 2023). As shown by Naserikia et al. (2023), LST and air temperature are strongly correlated, displaying similar

³<https://www.santepubliquefrance.fr/content/download/543105/3948412?version=1>

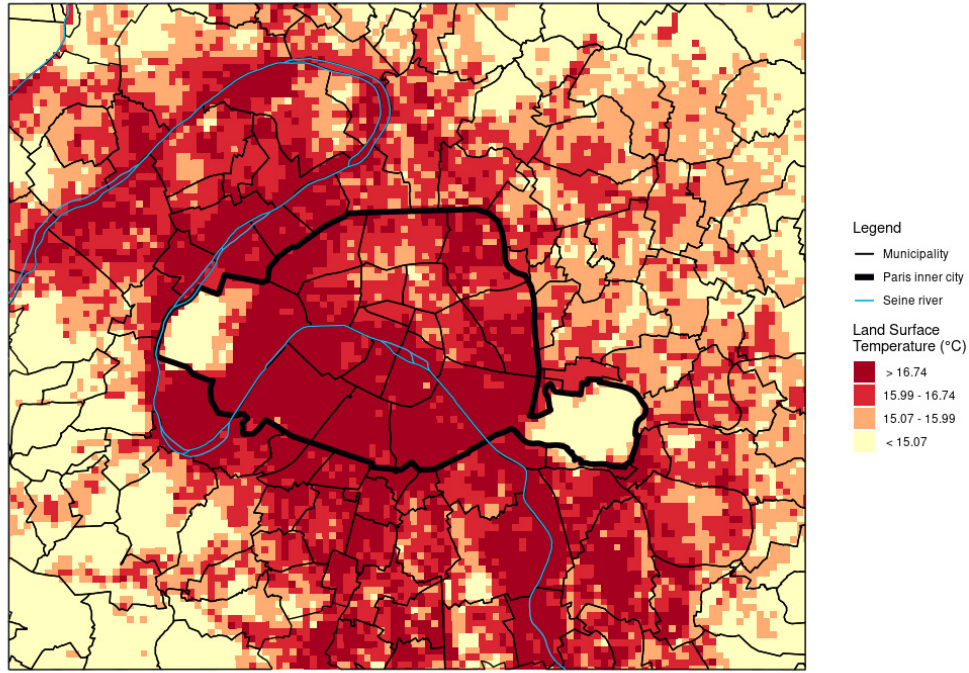


Figure 1: Land Surface Temperature from Thermocity data for Paris urban area (250 × 250 meters)

seasonal variations, albeit with differences in magnitude. This relationship establishes LST as a reliable proxy for air temperature. However, acquiring and processing LST data involves significant challenges. Weather conditions must not only reflect scenarios conducive to urban overheating but also meet the requirements for high-quality satellite observations, particularly with respect to cloud cover. Furthermore, the spatial resolution of the satellite data must be sufficient to accurately capture localized urban overheating. As a result, high-quality LST data are scarce, and their acquisition entails notable technological complexities.

In this context, the Thermocity project (Michel et al., 2022) offers a valuable dataset for analyzing urban overheating in France. The Thermocity project provides high resolution Land Surface Temperature data for the night of July 15, 2019. These measurements come from the ECOSTRESS satellite, a mission designed by the NASA to analyze the Earth Surface Temperature. The data are available in raster format with an extremely fine resolution of 70 meters. The data cover five French urban units: Marseille, Montpellier, Paris, Strasbourg, and Toulouse. Although the sample of urban units is relatively small, the Thermocity data encompass a substantial portion of the French population and represent diverse climate conditions. According to Köppen climate classification, Paris has an oceanic climate, featuring mild summers. Toulouse has a humid subtropical climate, leading to hot and humid summers. Montpellier and Marseille have a Mediterranean climate, with dry and hot summers. Specifically generated to analyze the urban heat island effect in these five urban areas, the LST data are particularly valuable for the present work. The data have been cleaned, consolidated by the Thermocity team, and are ready for analytical use. Figure 1 shows LST data for the Paris area. As depicted by the map, the center of Paris, very urbanized and densely populated, is the warmer part of the urban area, with numerous neighborhoods facing LST higher than 16.74°C that night. Conversely, open spaces and urban parks, such as *Bois de Boulogne* and *Bois de Vincennes*, respectively to the West and East of Paris inner city, feature low LST, below 15°C, illustrating the cooling effect of green zones.

The Thermocity project data provide a mean to quantitatively assess urban heat intensity using LST values, but they also enable the ranking of neighborhoods by comparing LST values to determine the relative intensity of urban overheating across each urban unit. An important concern regarding the urban overheating data is its validity for the entire temporal scope of this study. Indeed, Thermocity data are produced with one satellite observation, on July 15, 2019. The Thermocity data are designed to accurately represent urban overheating. Despite technological constraints,

Thermocity team managed to obtain satellite images for a summer night with perfect meteorological condition to observe urban overheating and urban heat islands. July 15, 2019 was an average warm sunny day, with exceptionally low cloud cover. These meteorological conditions were perfect to observe a representative Land Surface Temperature, that could be relevant for another time period and guide public decision of urbanization.

Despite the desire the Thermocity team has to provide robust data, two factors could potentially compromise the validity of the LST data: climate change and urban evolution. First, the evolution of the climate over the study period, increase the intensity of urban overheating over time. However, even though climate change can alter LST values over time, particularly by exacerbating them, it is expected that it will not affect the respective ranking of neighborhoods in terms of overheating intensity. Thus, climate change has only a minimal influence on the validity of the data in this context. Second, the evolution of the urban environment may also have influenced urban overheating over time. As detailed in section 3, urban overheating is partly determined by the architectural environment of the city. Thus, changes in the physical shape of the city over time may impact the validity of the Thermocity data. For example, the artificialization of a green space in favor of urbanized areas before July 2019 could lead to an overestimation of urban overheating in these neighborhoods in the Thermocity data. However, the examination of changes in land cover between 2012 and 2018 using Corine Land Cover raster shows that only 0.86% of the land in Marseille, Montpellier, Paris and Toulouse have changed their land cover classification between 2012 and 2018. This suggests that the physical form of the urban units in my sample has not significantly evolved, which supports the validity of the LST data. Finally, as discussed in Section 5, the identification strategy relies on the use of an instrumental variable. This instrumental variable extracts variations in LST that are not specific to the night of July 19, 2019. As a result, it eliminates a potential measurement error bias associated with relying on a single observation of LST (see Section 7.1 for more details).

4.2 Housing data

The housing data used in this study is sourced from the *Direction Générale des Finances Publiques* through Cerema, known as the DV3F database.⁴ This administrative database contains information on all property transfers made by notarial deed since 2010, covering the entirety of France except the Alsace and Lorraine regions. The DV3F database includes the precise location of the transacted dwellings in the form of XY coordinates, as well as information on the physical characteristics of the dwellings. The raw database contains over 10 million observations for the period 2010-2023. This database is widely used to study housing in France (see for example: Belgodere and Casamatta, 2023; Chapelle et al., 2023; Chapelle and Eymeoud, 2022; Coinon et al., 2023)

The study area of this work is limited by the urban units covered by both the Thermocity project and the DV3F data. Since Strasbourg (in Alsace) is not included in the DV3F data, the analysis focuses on the four other urban units: Paris, Marseille, Toulouse, and Montpellier. Furthermore, only data from 2013 to 2019 are used. Data after 2019 are excluded due to recovery biases, particularly related to the COVID-19 pandemic, which disrupted real estate activities in France, as well as the lack of consolidated data for 2022, 2023, and 2024. LST data were produced in 2019. I use the 2013–2019 period for the analysis, assuming that the urban environment was not profoundly transformed over these 7 years. As a result of this temporal and spatial scope, the database contains 1 124 127 observations.

The DV3F database contains some noise, originating from atypical property transfers, such as combinations of multiple dwellings. To ensure data accuracy, only property transfers involving a single apartment or house are retained in the database. Dwellings with missing information, such as the number of rooms or the construction period, are also removed from the data. Finally, outliers are removed from the database. The definition of outliers is akin to a winsorization procedure, following Puspapuram et al. (2017). This process involves employing a basic hedonic price model and eliminating observations with residuals in the top or bottom 2.5%. Next, I exclude dwellings with a floor space of less than 9m², as these do not comply with the minimum floor space standards applicable in France, and with a selling price lower than €45 000, as these transactions might represent particular transactions with prices not being the outcome of the housing market, like transfers between family members. The final database contains 973 645 observations of housing transactions.

⁴Données foncières retraitées par le Cerema - DGALN DGFIP, see : <https://datafoncier.cerema.fr/donnees/donnee-dv3f>

4.3 Construction of the database

To complete the analysis, the housing data must be enriched with additional spatial information: LST data, density data, land cover data, and wind speed data. The sources of these datasets and their specific treatments are detailed in the following paragraph. Given that each dataset has a unique spatial resolution, I harmonize the scales by aggregating the data into a grid of 250×250 -meter squares. For each square, the average value of each variable is calculated. These squares are referred to as Reference Spatial Units (hereafter RSUs) throughout the remainder of this article. The 250×250 -meter scale is chosen to represent the close environment of each dwelling while maintaining a certain level of granularity. Using RSUs also facilitates the integration of land cover data. Each property is thus associated with the RSU in which it is located, and the spatial data contained within that RSU.

The average LST per RSU is computed based on data from the Thermocity project. Density data is provided by the *Carreaux* dataset from INSEE. These are gridded data with a resolution of 200 meters, covering all of France and containing various socio-economic information. The dataset includes a variable indicating the number of residents in each grid cell, which is used to measure population density. The average density for each RSU is then calculated. Land cover data is sourced from the 2018 Corine Land Cover database, which provides detailed Shapefiles covering the entirety of France with a precision of 20 meters. The original database includes 42 land cover categories. To simplify the interpretation and processing of these data, the 42 categories are grouped into 5 broad classes based on the Corine Land Cover classification: urban land, industrial land, green spaces within urban land, water, and natural land, which encompasses agricultural, wetland, and natural zones. For each RSU, five variables representing the proportion of the RSU covered by a specific type of land are computed. Finally, the average wind speed per RSU is calculated using data from the Global Wind Atlas, which provides global wind speed data at a resolution of 200×300 meters, generated through meteorological simulations.

Second, location information is incorporated. IRIS (Ilots Regroupés pour l'Information Statistique) divisions are added to the dataset. IRIS are statistical units defined by INSEE, typically corresponding to large neighborhoods with an average population of about 2000 inhabitants. Each housing unit is linked to the IRIS code of the division it belongs to. Additionally, the municipality in which each dwelling is located is included in the database. The municipality is the intermediate administrative level between the urban unit and the IRIS. Using the precise geographic coordinates of the dwellings, the straight-line distance from each observation to the center of its corresponding urban area is calculated.⁵ Finally, the average density per IRIS is computed based on the average density per RSU, and the RSU relative density to the average IRIS density is then calculated to better account for the variability in population density within each IRIS.

Table 1 presents descriptive statistics on the entire estimation dataset and by urban unit. These descriptive statistics aim to provide insights into the different situations regarding housing prices and LST per RSU in the sample. As expected, dwellings in the urban unit of Paris represent 74.2% of the total sample. It is also the most expensive urban unit, with an average price of €338952. Toulouse and Montpellier are the two least expensive urban areas in the sample, with an average price around €200000. Regarding the average LST, as expected, the northern urban unit, Paris, exhibits the coolest average LST with a mean of 16.08°C. Montpellier is the hottest urban unit, with an average LST of 24.20°C. The standard deviation indicates that Montpellier exhibits the least dispersion in the sample, reflecting the city's homogeneity and the lack of surrounding relief. Although both Montpellier and Marseille are Mediterranean urban units, thus exposed to the same climate and a comparative amount of sunshine, the distribution of LST varies significantly, indicating the influence of the city's physical characteristics on LST. In absolute, the LST values do not seem extreme, and may even suggest a rather temperate climate. However, the reality is quite different. Indeed, this table clearly illustrates how much heat the urban environment absorbs. For each city, the surface temperatures near the minimum values reflect those of the rural surroundings around the city. For Paris, this temperature is around 9°C. The average LST is 8°C higher. This temperature gap can be interpreted as the amount of heat retained by the urban environment. This brief examination demonstrates that the LST values in the city centers are indeed structurally high in comparison to those of the surrounding rural areas.

⁵City centers are defined using Google Maps queries.

Table 1: Descriptive statistics for housing price and Land Surface Temperature per RSU

	Min	1st Qu.	Median	Mean	3rd Qu.	Max	StD
Housing Price							
All N = 973 645	45 016	160 000	243 000	306 604	370 000	2 990 000	238 368
Paris N = 722 739	45 190	180 000	266 000	338 952	405 000	2 990 000	256 893
Marseille N = 132 898	45 016	123 000	190 000	227 239	290 000	1 800 000	148 793
Toulouse N = 85 845	45 380	110 200	165 000	194 887	249 000	1 467 420	115 217
Montpellier N = 32 163	45 200	109 600	170 000	205 811	270 000	1 153 700	130 996
Land Surface Temperature per RSU							
All	8.853	15.76	16.70	17.55	18.36	26.22	2.859
Paris	8.853	15.40	16.24	16.08	16.94	19.72	1.165
Marseille	13.11	20.34	21.45	21.14	22.15	24.86	1.438
Toulouse	13.33	20.75	22.42	21.90	23.48	25.05	1.979
Montpellier	17.74	23.64	24.40	24.20	24.95	26.22	1.137

Note: Housing prices are expressed in euros

5 Empirical Strategy

This section is devoted to the empirical strategy employed to identify the effect of urban overheating on housing prices. Subsection 5.1 presents the basic hedonic price framework and explains why, within this framework, an OLS estimation of the effect of urban overheating on housing prices might be biased. Subsection 5.2 presents the instrumental variable approach designed to achieve an unbiased estimation of urban overheating effect.

5.1 Hedonic price framework

As seen previously, hedonic pricing model are widely used to analyze the impact of different factors on housing prices. These models assume that the price of a dwelling perfectly reflects the implicit price of all its intrinsic characteristics and that of its location. By including the urban overheating variable in the model, it is possible to estimate the effect of urban overheating on housing prices. The hedonic price model estimated here can be written as follows:

$$\log(P_{itn}) = \beta_0 + \beta_1 \times UO_i + \beta_2 X_i + \beta_3 Z_i + \delta_t + \delta_n + \varepsilon_{itn} \quad (1)$$

with P_{itn} the price of the dwelling i sold in year t , located in the IRIS n , UO_i the measure of the RSU urban overheating, X_i a vector of dwelling's physical characteristics, Z_i a vector of control variables for the location of the dwelling i , δ_t year fixed-effects, δ_n IRIS fixed-effects, and ε_{itn} an error term independent and identically distributed.

The vector of dwelling's physical characteristics contains a set of dummy variables for the number of rooms, the type of dwelling (house or apartment), the number of outbuildings (e.g. garage, shed, cellar...), and a set of dummy variables for the period of construction. The vector of location's attributes consists in the density relative to the average density of the IRIS, and the bird's-eye distance of the dwelling to the center of the urban unit. Year fixed-effects control for all variations in housing prices common to a year. IRIS fixed-effects control for all determinants of housing prices common to a neighborhood. For the remainder of the analysis, standard errors are clustered at the lowest level, the RSU level.

Since IRIS are fine-scale administrative division, this implies that identification hinges on the intra-IRIS variations of the variable of interest, measured at a smaller scale, the RSU level. Questions may arise regarding the number of RSUs per IRIS, as an insufficient number could fail to ensure sufficient variability in LST within the IRIS. On average,

each IRIS contains 5 RSUs, which seems sufficient to capture variations in LST within the IRIS. Some IRIS consist of a single RSU. However, these account for just over 9000 observations, representing less than 1% of the sample. Quantitatively, there are significant intra-IRIS variations in the variables of interest. The average intra-IRIS coefficients of variation for price per square meter and land surface temperature are 0.452 and 0.031 respectively. Variations in price per square meter could be due to unobserved fine-scale characteristics, which could lead to endogeneity if correlated with urban overheating. The coefficient of variation for LST is lower, but still suggests that there is sufficient intra-IRIS variation in urban overheating.

Endogeneity may be due to omitted variables and/or measurement error. Various unobserved factors affecting housing prices may be correlated with urban overheating. As mentioned above, urban vegetation, such as small parks or urban trees, could be an important confounding factor. In addition to its cooling potential, vegetation provides numerous benefits to neighborhood residents that are likely to positively affect housing prices. In particular, vegetated areas have aesthetic qualities and play an important social role by providing opportunities for walking and recreation. In this context, the omission of fine-scale vegetation could lead to an underestimation of the impact of urban overheating on housing prices. Additionally, unobserved intra-IRIS socioeconomic factors, such as household wealth, might be another source of endogeneity. Households might sort themselves in such a way that wealthier households live in neighborhoods with better thermal infrastructures and more green spaces, resulting in cooler temperatures and higher housing prices, while poorer households may be confined to degraded neighborhoods with fewer green spaces and, consequently, higher temperatures. In Paris, for example, the neighborhoods around the Boulogne Woods are among the most expensive and also host the wealthiest population in the city. Socio-economic characteristics vary significantly within IRIS. Using *Carreaux* data from INSEE, the average intra-IRIS coefficient of variation can be calculated for the share of poor households, the sum of living standards, and the share of owner-occupied households, each calculated at the RSU level. These are 0.282, 0.276 and 0.194 respectively, suggesting strong local variation in the socio-economic composition of the IRIS, which could potentially cause endogeneity if correlated with urban overheating. Unobserved building characteristics could also introduce endogeneity, as they could contribute to increasing housing prices while reducing the sensitivity of dwellings to urban overheating. In addition, the measure of urban overheating used here may also be subject to error. In fact, the LST is not a direct measure of air temperature and therefore does not fully capture the extent of urban overheating. Additionally, the LST data were collected on a specific day, which, although intended to be representative of typical conditions, might introduce measurement errors. These errors could lead to regression attenuation, resulting in an underestimation of the effect of urban overheating on housing prices when using OLS (Frost and Thompson, 2000).

To determine whether the LST variable is indeed endogenous, the Durbin–Wu–Hausman test (Durbin, 1954; Wu, 1973; Hausman, 1978) have been conducted to test for endogeneity. If there is a significant difference between OLS and 2SLS estimations, it indicates that the regressor variable is endogenous, necessitating the use of instrumental variables. The p-value of the test is below 0.05, allowing us to reject the null hypothesis and confirm that the model does indeed exhibit endogeneity, making 2SLS estimation appropriate in this case.

Given the issue of endogeneity, I develop an instrumental variable approach, presented in the remainder of this section, which addresses the endogeneity arising from unobserved variables and measurement errors.

5.2 Instrumental Variable

In this subsection, I present the instrumental variable framework. Wind speed, aggregated at the RSU scale ($250\text{m} \times 250\text{m}$), is chosen as an instrument. In section 5.2.1, a precise description of the instrument is provided. The other two sections are dedicated to verifying and discussing the hypotheses that this instrument must satisfy, namely relevance in section 5.2.2 and exogeneity in section 5.2.3.

5.2.1 Description of the instrument

The wind speed data are sourced from the Global Wind Atlas dataset spans across France, providing simulated wind speed data in meters per second at a height of 10 meters above ground level, with a spatial resolution of 200×300 meters. These data are derived from meteorological simulations that combine a mesoscale and a microscale models to estimate average wind speeds for a typical year. The mesoscale model captures large-scale meteorological condi-

tions, calibrated using daily data from the European Centre for Medium-Range Weather Forecasts covering the period 2008-2017. This calibration ensures that the input data accurately represents average meteorological conditions for the 2008-2017 period. The microscale model accounts for fine-scale natural topography and surface roughness. The surface roughness index classifies the terrain into four categories ordered by decreasing roughness: forests, urban areas, plains, and water bodies. A higher roughness index generally correlates with lower wind speeds. The interplay between the landscape features and the meteorological models results in spatially heterogeneous wind speed patterns within the urban areas and the IRIS. For each RSU grid, the average wind speed for each RSU is computed. In the following, this variable is referred to as the average wind speed.

Table 2 presents descriptive statistics for the instrument across the entire sample and city samples. The distribution of average wind speed varies significantly from one city to another. Marseille has the highest observed wind speeds and the most dispersed distribution, notably due to relief. Paris and Montpellier have the lowest dispersion. Montpellier also has the lowest maximum wind speed among the urban units. Finally, the average intra-IRIS coefficient of variation for wind speed is 0.053, indicating that there is significant variation in wind speed within the IRIS.

Table 2: Descriptive statistics for the Wind Speed per RSU

	Min	1st Qu.	Median	Mean	3rd Qu.	Max	StD
Average Wind Speed per RSU (m/s)							
All	0.720	2.325	2.461	2.572	2.719	6.865	0.393
Paris	1.370	2.304	2.418	2.498	2.627	5.032	0.295
Marseille	0.720	2.405	2.669	2.805	3.076	6.865	0.613
Toulouse	2.166	2.483	2.682	2.814	2.989	6.038	0.456
Montpellier	1.984	2.406	2.527	2.622	2.751	4.808	0.327

5.2.2 Instruments Relevance

The reason for employing the average wind speed lies in the fact that wind speed is one of the factors that generally contribute to cool the environment. However, urban morphology obstructs and slows down wind flows, resulting in less effective cooling in urban areas compared to rural areas. Therefore, areas with low wind speeds are more susceptible to urban overheating and areas with high wind speeds are less susceptible to urban overheating.

To quantify the relevance of the instrument, it is possible to examine the results of the first stage of the 2SLS estimation of the model. Table 3 presents the first-stage results of the 2SLS estimation of the effect of urban overheating on housing prices, along with the associated F-Stats for the IV. To account for heteroscedasticity stemming from large sample, the effective F-Stat from Luis et al. (2013) is used. Due to space constraints, only the coefficients related to the instruments are presented here. As expected, the estimated coefficient for wind speed is negative and statistically different from 0, confirming with the fact that wind speed cools the environment. Furthermore, the F-Stat associated with the model is far above 10, the rule of thumb, suggesting that the instrument is strong and very relevant for the analysis.

5.2.3 Instrument Exogeneity

The second condition that the instrument must meet is the exogeneity condition. Here, this means that, conditionally on all control variables, the instruments should have no impact on housing prices other than through the urban overheating variable, and in particular not be correlated to unobserved determinants of housing prices, that is, factors that drive households' location choices. The next paragraphs discuss why the instrument presented above might fulfill this condition.

Wind speed can verify the exogeneity condition, as it is determined by nature and cannot be directly controlled by humans. However, it is necessary to question the link between wind speed and fine-scale, that is intra-IRIS, architectural shape of the city, since the wind speed is adversely affected by urban construction, as buildings hinder airflow. Thus, denser neighborhoods within IRIS, which are also the most expensive, might have the slowest wind. However,

Table 3: First stage results

Land Surface Temperature	
Wind Speed	−0.467*** (0.000)
<i>Control variables:</i>	
Dwelling characteristics	Yes
Location attributes	Yes
<i>Fixed-effects:</i>	
IRIS FE	Yes
Year FE	Yes
Num. Obs.	973 645
R2	0.978
F-Stat	861.47

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered by RSU; dwelling characteristics contain the number of rooms, the type of the dwelling (house or flat), the number of outbuildings and the period of construction; location's attributes contain the relative density and the distance to center; Land Surface Temperature and Wind Speed are aggregated at RSU level.

this relation seems unlikely to happen with this dataset, since, as the landscape roughness, but not the precise urban form, is taken into account to generate wind speed data.

Beyond the relationship with the urban physical form, it is possible to question whether wind can be considered as an (dis)amenity so that it might influence household location choices. Though air pollution, wind has played a determining role in spatial sorting by income in many French cities, especially in Paris and Marseille. However, it does not operate at the within-IRIS scale, and cannot therefore generate any correlation with unobservable. Wind might also cause noise pollution or simply discomfort for residents. However, the wind speeds observed in this sample, based on averages from the 2008–2017 period, are unlikely to be perceived negatively. Regional winds, such as the Mistral in Marseille, often viewed unfavorably, typically blow at an average speed of 13 m/s, which is significantly stronger than the values observed in this sample.⁶ Moreover, regional winds operate on a much larger scale than the IRIS. Lastly, the winds in my sample do not indicate exposure to storm risks, which also occur at a much larger scale than IRIS.

6 Results

Table 4 presents the results of the OLS and the 2SLS estimation of the effect of overheating on housing prices. Column (1) shows the results obtained with OLS and the entire dwelling sample. Column (2) shows the results obtained with 2SLS. Since urban overheating shows significant variation across urban units due to differences in local climate conditions and urbanization, results separating the sample by urban units are also shown in the Table 4. Specifically, column PRS shows the results for Paris, column MSL, TLS and MPL, for Marseille, Toulouse, and Montpellier, respectively. In order to assess differences in the impact of urban overheating over time, Table 4 also displays the results of the 2SLS estimations for two different time-periods: the 2013–15 period and the 2016–2019 period. The division into two periods is guided by two criteria. Firstly, 2015 holds significance in France due to experiencing a major heatwave. Secondly, all years following 2015 are marked by significant summer heatwaves. Thus, the effect of urban overheating is expected to increase after 2015 due to two potentially cumulative factors: the potential accentuation of urban overheating and increased household awareness of urban overheating. Column 2013–15 presents the results for the years 2013, 2014, and 2015, while column 2016–19 presents the results for the years 2016, 2017, 2018, and 2019.

⁶<https://meteofrance.com/comprendre-la-meteo/le-vent/les-vents-regionaux>

Table 4: OLS and 2SLS estimates of the urban overheating effect on housing prices

	Logarithm of selling price							
	OLS		2SLS					
	FULL (1)	FULL (2)	PRS (3)	MSL (4)	TLS (5)	MPL (6)	2013-15 (7)	2016-19 (8)
UO effect	-0.009*** (0.000)	-0.042*** (0.000)	-0.025** (0.041)	-0.102*** (0.000)	-0.021*** (0.003)	-0.004 (0.964)	-0.039*** (0.000)	-0.043*** (0.000)
Distance to center (km)	0.006*** (0.005)	0.012*** (0.000)	0.010*** (0.007)	-0.001 (0.850)	0.018*** (0.000)	0.018 (0.540)	0.013*** (0.000)	0.011*** (0.000)
Relative density	-0.023*** (0.000)	-0.020*** (0.000)	-0.022*** (0.000)	-0.016*** (0.000)	-0.016*** (0.000)	-0.036*** (0.000)	-0.019*** (0.000)	-0.020*** (0.000)
House dummy	0.511*** (0.000)	0.508*** (0.000)	0.509*** (0.000)	0.495*** (0.000)	0.504*** (0.000)	0.446*** (0.000)	0.485*** (0.000)	0.522*** (0.000)
Number of dependencies	0.125*** (0.000)	0.126*** (0.000)	0.135*** (0.000)	0.136*** (0.000)	0.074*** (0.000)	0.064*** (0.000)	0.130*** (0.000)	0.124*** (0.000)
Dwelling with one room	REF	REF	REF	REF	REF	REF	REF	REF
Dwelling with 2 rooms	0.463*** (0.000)	0.463*** (0.000)	0.482*** (0.000)	0.378*** (0.000)	0.416*** (0.000)	0.406*** (0.000)	0.461*** (0.000)	0.465*** (0.000)
Dwelling with 3 rooms	0.799*** (0.000)	0.799*** (0.000)	0.817*** (0.000)	0.697*** (0.000)	0.766*** (0.000)	0.764*** (0.000)	0.799*** (0.000)	0.799*** (0.000)
Dwelling with 4 rooms	0.984*** (0.000)	0.983*** (0.000)	1.004*** (0.000)	0.870*** (0.000)	0.954*** (0.000)	0.935*** (0.000)	0.984*** (0.000)	0.982*** (0.000)
Dwelling with 5 rooms	1.176*** (0.000)	1.174*** (0.000)	1.194*** (0.000)	1.043*** (0.000)	1.176*** (0.000)	1.142*** (0.000)	1.174*** (0.000)	1.175*** (0.000)
<i>Other control variables:</i>								
Construction Period	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fixed-effects:</i>								
IRIS FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Num.Obs.	973 645	973 645	722 739	132 898	85 845	32 163	366 455	607 190
R2	0.822	0.821	0.813	0.784	0.776	0.795	0.819	0.827
F-Stat		861.47	345.94	210.70	448.68	13.64	744.80	835.19

* p < 0.1, ** p < 0.05, *** p < 0.01

Note: UO stands for urban overheating; standard errors are clustered by RSU; instrument used is the average wind speed; PRS stands for Paris; MSL for Marseille; TLS for Toulouse; MPL for Montpellier; 2013-15 stands for dwellings sold before January the 1st, 2016; 2016-19 for dwellings sold after January the 1st, 2016; Relative density is the density relative to the average density of the IRIS; House dummy is a dummy variable taking 1 if the dwelling a house, 0 otherwise.

Both OLS and 2SLS estimations point to a negative impact of urban overheating on housing price, indicating the anticipated detrimental effects of urban overheating. Additionally, estimated coefficients are statistically significant. The results obtained with OLS predict a small effects of urban overheating, with an estimated coefficient of -0.009 . The estimated coefficient increases when accounting for the endogeneity of urban overheating, indicating that a 1-degree increase in average urban overheating is associated with a 4.2% decrease in housing prices. The estimated coefficients for control variables are consistent across OLS and 2SLS regressions. Comparing the results (1) and (2) highlights the bias inherent in OLS estimation, which here underestimates the effect of urban overheating on housing prices. As detailed in the previous section, this bias may arise from omitted variables or measurement errors. A confounding factor that would produce this bias should be positively associated with both urban overheating and housing prices. However, the confounding factors previously presented, such as unobserved physical characteristics of buildings, are positively associated with overheating but negatively with housing prices, which should lead to an OLS bias in the opposite direction. Therefore, it is more likely that measurement errors are the source of the bias, precisely an attenuation bias that leads to an underestimation of the effect of urban overheating using OLS (see Section 7.1 for more details).

For Paris, the 2SLS estimated coefficient is -0.025 and is statistically significant, meaning that a 1-degree increase in urban overheating is associated with a 2.5% decrease in housing prices. For Toulouse, the estimation predicts a similar effect of urban overheating on housing price, with a coefficient of -0.021 . The F-Stats are also above the threshold of 10, at 345.94 for Paris and 448.68 for Toulouse, respectively, confirming the strength of the instruments for these two urban units. For Montpellier, the estimated coefficient is -0.004 , but it is not statistically different from 0. The F-statistic is 13.64, which, although above the rule of thumb threshold of 10, is relatively low. To make sense of this result, it needs to be contextualized alongside the descriptive statistics provided in Table 1. The descriptive statistics point to the low variability of LST in the Montpellier urban unit, with a standard deviation of 1.138, the lowest in the sample. This suggests that while Montpellier experiences high temperatures, the temperature differential between the hottest and coolest neighborhoods is not substantial. That can be linked to the real geography of Montpellier, which is very flat and homogenous. Consequently, according to the urban overheating characterization used in this analysis, housing prices in Montpellier do not seem to be significantly affected by urban overheating. For Marseille, the estimated effect is rather high, at -0.098 , suggesting that a 1-degree increase in average LST would be associated with a 9.8% decrease in housing prices. This result is surprising, especially considering that the F-statistic, at 211.45, confirms the strength of the instrument. This surprising result likely stems from the unique characteristics of Marseille's urban unit. The challenge lies in capturing all local amenities, which may operate at finer scales than the IRIS. Bono et al. (2008) shows the importance of highly localized factors in housing prices formation in Marseille. Among these localized amenities, building characteristics can influence both housing prices and urban overheating, while escaping the IRIS fixed-effects.

The estimated coefficient for the 2013-2015 period is -0.039 . The F-statistic is 744.80, confirming the strength of the instrument. The estimated coefficient for the 2016-2019 period points to a decrease by 4.3% in housing prices for a 1-degree increase in LST, that is an increase in the impact by about 10% as compared to the first period. Since coefficients are very precisely estimated, this difference is statistically significant. Similarly, the F-statistic is very high (835.19), confirming the strength of the instrument for the second period. The estimated effect of urban overheating on housing prices is higher for the first period than for the second period. This result confirms the hypothesis stated earlier: after 2015, households preferences are more sensitive to urban overheating and successive heatwaves and the acceleration of climate change lead to an intensification of the detrimental impact of urban overheating on housing prices. However, as seen in section 4.1, urban overheating might not have changed that much on the 2013-2019 period. This suggests that the change is due to households being more sensitive to urban overheating at the end of the period than at the beginning.

7 Robustness checks

In this section, several robustness checks are implemented. Section 7.1 presents simulations to show that the instrument variable strategy is able to remove dilution bias stemming from measurement errors. Section 7.2 presents 2SLS estimation with alternative specifications, including land cover variables and location $\times$ year fixed-effects. Section 7.3 provides evidences on the validity of the LST datasets using another urban overheating dataset.

7.1 Measurement errors

A key strength of the empirical strategy is its ability to account for potential biases in OLS estimates caused by measurement errors, due to the fact that LST is a proxy for air temperature. To verify whether the instrumental variable effectively addresses this issue, simulations are performed bellow, following the procedure used by Exbrayat and Stephane (2024). A noisy version of the LST variable, denoted as $L\hat{S}T$, is created by adding a random term drawn from a normal distribution with a mean of 0 and different values for the standard deviation, denoted τ . This approach allows us to examine how increasing measurement error affects the estimated impact of urban overheating on housing prices by OLS and 2SLS. The results are presented in Table 5. The OLS estimates are indeed subject to attenuation bias, as an increase in the noise in the overheating variable leads to a reduction in the predicted effect. In contrast, the 2SLS estimates using wind speed as instrumental variable remain unchanged, regardless of the level of noise introduced. This contributes to show that the instrumental variable method employed is consistent and effectively eliminates attenuation bias caused by measurement errors in the urban overheating variable.

Table 5: OLS and 2SLS estimates of the urban overheating effect on housing prices with noisy Land Surface Temperature

	Logarithm of selling price					
	OLS			2SLS		
	$\tau = 0.1$	$\tau = 0.5$	$\tau = 1$	$\tau = 0.1$	$\tau = 0.5$	$\tau = 1$
$L\hat{S}T$	-0.009*** (0.000)	-0.004*** (0.000)	-0.001*** (0.000)	-0.042*** (0.000)	-0.041*** (0.000)	-0.042*** (0.000)
<i>Control variables:</i>						
Dwelling characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Location attributes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fixed-effects:</i>						
IRIS FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Num.Obs.	973 645	973 645	973 645	973 645	973 645	973 645
R2	0.821	0.821	0.821	0.820	0.820	0.816
F-Stat				858.52	845.70	762.22

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: $L\hat{S}T$ stands for noisy Land Surface Temperature variable; standard errors are clustered by RSU; instrument used is the average wind speed; dwelling characteristics contain the number of rooms, the type of dwelling (house or flat), the number of outbuildings and the period of construction; location's attributes contain the density relative to IRIS mean and the distance to center.

7.2 Including land cover control variable, Municipality $\times$ Year and IRIS $\times$ Year fixed-effects

Land cover might have huge impact on both housing prices and urban overheating. Especially, the presence of green spaces in the RSU cools the environment and increases housing prices due to aesthetic features. The use of land cover variables computed with Corine Land Cover, allows for a better consideration of the environment in which the dwelling is located, especially on local vegetation. The idea is therefore to observe if a significant effect of urban overheating

persists after the introduction of these variables.

The table 6 presents the results obtained from estimations of the effect of urban overheating on housing prices via OLS and 2SLS with land cover as additional control variables. The estimated coefficient for the instrument in the first stage is reduced by the introduction of land cover: it changes from -0.467 to -0.347 . These changes are likely due to some correlation between the instrument and the land cover variables. As expected, urban and industrial land covers have a weaker mitigating effect on urban overheating as compared to parks and natural land covers, which illustrate the cooling benefits of green and natural spaces. For both the OLS and 2SLS estimations, the predicted effect of urban overheating on house prices is greater with the additional controls. For OLS, the coefficient changes from -0.009 to -0.012 . For 2SLS estimation, the estimated effect of urban overheating increases, changing from -0.042 to -0.060 . Since coefficients are very precisely estimated, differences between coefficients are statistically significant. The estimated coefficients for land cover are all statistically different from 0. The F-statistic in Table 6 is above the threshold of 10, but is lower than in the estimation without land cover.

The inclusion of Municipality/IRIS $\times$ Year fixed effects allows controlling for trends specific to municipalities/IRIS, such as the rapid gentrification occurring in certain districts of Marseille. Table 7 presents the 2SLS estimation results for the effects of urban overheating on housing prices, including Municipality $\times$ Year fixed effects, while the table 8 presents the 2SLS estimation results including IRIS $\times$ Year fixed effects. In both tables, estimated coefficients are very similar to those from the previous section. For instance, the estimated coefficient for Paris changes only slightly, from -0.025 to -0.024 . Moreover, while one might have expected the estimated effect for Marseille to vary, particularly given that some of the city's disadvantaged neighborhoods underwent urban renewal during the study period, hence changing the dynamics of housing prices, the estimated coefficient shows minimal change. Similarly, F-stats do not change significantly.

Table 6: OLS and 2SLS estimates of the urban overheating effect on housing prices with land cover control variables

	Logarithm of selling price	
	OLS	2SLS
<i>First step:</i>		
Average Wind Speed		−0.347*** (0.000)
Water Land		REF
Urban Land		−1.259*** (0.000)
Park land		−2.012*** (0.000)
Industrial Land		−1.130*** (0.000)
Nature Land		−2.629*** (0.000)
UO effect	−0.012*** (0.000)	−0.060*** (0.000)
Water land	REF	REF
Urban land	−0.089*** (0.000)	−0.148*** (0.000)
Park land	−0.106*** (0.000)	−0.203*** (0.000)
Industrial land	−0.108*** (0.000)	−0.161*** (0.000)
Nature land	−0.117*** (0.000)	−0.246*** (0.000)
<i>Other control variables:</i>		
Dwelling characteristics	Yes	Yes
Location attributes	Yes	Yes
<i>Fixed-effects:</i>		
IRIS FE	Yes	Yes
Year FE	Yes	Yes
Num.Obs.	973 645	973 645
R2	0.822	0.821
F-Stat		861.46

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: UO stands for urban overheating; standard errors are clustered by RSU; instrument used is the average wind speed; dwelling characteristics contain the number of rooms, the type of dwelling (house or flat), the number of outbuildings and the period of construction; location's attributes contain the relative density and the distance to center.

Table 7: OLS and 2SLS estimates of the urban overheating effect on housing prices with Municipality $\times$ Year fixed-effects

	Logarithm of selling price							
	OLS		2SLS					
	FULL (1)	FULL (2)	PRS (3)	MSL (4)	TLS (5)	MPL (6)	2013-15 (7)	2016-19 (8)
UO effect	-0.009*** (0.000)	-0.041*** (0.000)	-0.024* (0.051)	-0.102*** (0.000)	-0.020*** (0.004)	-0.009 (0.911)	-0.039*** (0.000)	-0.042*** (0.000)
Distance to center (km)	0.006*** (0.003)	0.012*** (0.000)	0.011*** (0.004)	-0.001 (0.898)	0.018*** (0.000)	0.020 (0.483)	0.013*** (0.000)	0.012*** (0.000)
Relative density	-0.023*** (0.000)	-0.020*** (0.000)	-0.023*** (0.000)	-0.016*** (0.000)	-0.016*** (0.000)	-0.036*** (0.000)	-0.019*** (0.000)	-0.021*** (0.000)
House dummy	0.511*** (0.000)	0.508*** (0.000)	0.509*** (0.000)	0.495*** (0.000)	0.505*** (0.000)	0.445*** (0.000)	0.486*** (0.000)	0.522*** (0.000)
Number of dependencies	0.126*** (0.000)	0.126*** (0.000)	0.136*** (0.000)	0.136*** (0.000)	0.075*** (0.000)	0.063*** (0.000)	0.130*** (0.000)	0.124*** (0.000)
Dwelling with one room	REF	REF	REF	REF	REF	REF	REF	REF
Dwelling with 2 rooms	0.464*** (0.000)	0.464*** (0.000)	0.482*** (0.000)	0.378*** (0.000)	0.417*** (0.000)	0.407*** (0.000)	0.461*** (0.000)	0.465*** (0.000)
Dwelling with 3 rooms	0.800*** (0.000)	0.799*** (0.000)	0.818*** (0.000)	0.696*** (0.000)	0.766*** (0.000)	0.763*** (0.000)	0.799*** (0.000)	0.799*** (0.000)
Dwelling with 4 rooms	0.984*** (0.000)	0.984*** (0.000)	1.005*** (0.000)	0.871*** (0.000)	0.954*** (0.000)	0.935*** (0.000)	0.985*** (0.000)	0.982*** (0.000)
Dwelling with 5 rooms	1.178*** (0.000)	1.176*** (0.000)	1.195*** (0.000)	1.044*** (0.000)	1.177*** (0.000)	1.142*** (0.000)	1.175*** (0.000)	1.176*** (0.000)
<i>Other control variables:</i>								
Construction Period	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fixed-effects:</i>								
IRIS FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipality $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Num.Obs.	973 645	973 645	722 739	132 898	85 845	32 163	366 455	607 190
R2	0.825	0.824	0.816	0.793	0.779	0.797	0.820	0.828
F-Stat		863.54	346.44	211.89	451.93	14.20	739.02	836.86

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: UO stands for urban overheating; standard errors are clustered by RSU; instrument used is the average wind speed; PRS stands for Paris; MSL for Marseille; TLS for Toulouse; MPL for Montpellier; 2013-15 stands for dwellings sold before January the 1st, 2016; 2016-19 for dwellings sold after January the 1st, 2016; Relative density is the density relative to the average density of the IRIS; House dummy is a dummy variable taking 1 if the dwelling a house, 0 otherwise.

Table 8: OLS and 2SLS estimates of the urban overheating effect on housing prices with IRIS $\times$ Year fixed-effects

	Logarithm of selling price							
	OLS		2SLS					
	FULL (1)	FULL (2)	PRS (3)	MSL (4)	TLS (5)	MPL (6)	2013-15 (7)	2016-19 (8)
UO effect	-0.009*** (0.000)	-0.041*** (0.000)	-0.024** (0.046)	-0.102*** (0.000)	-0.019*** (0.003)	-0.012 (0.888)	-0.038*** (0.000)	-0.043*** (0.000)
Distance to center (km)	0.006*** (0.004)	0.012*** (0.000)	0.010*** (0.005)	-0.001 (0.895)	0.018*** (0.000)	0.021 (0.482)	0.013*** (0.000)	0.012*** (0.000)
Relative density	-0.023*** (0.000)	-0.020*** (0.000)	-0.023*** (0.000)	-0.016*** (0.000)	-0.017*** (0.000)	-0.035*** (0.000)	-0.019*** (0.000)	-0.020*** (0.000)
House dummy	0.510*** (0.000)	0.508*** (0.000)	0.509*** (0.000)	0.495*** (0.000)	0.503*** (0.000)	0.445*** (0.000)	0.486*** (0.000)	0.521*** (0.000)
Number of dependencies	0.126*** (0.000)	0.126*** (0.000)	0.136*** (0.000)	0.136*** (0.000)	0.074*** (0.000)	0.065*** (0.000)	0.130*** (0.000)	0.124*** (0.000)
Dwelling with one room	REF	REF	REF	REF	REF	REF	REF	REF
Dwelling with 2 rooms	0.464*** (0.000)	0.463*** (0.000)	0.482*** (0.000)	0.378*** (0.000)	0.415*** (0.000)	0.406*** (0.000)	0.461*** (0.000)	0.465*** (0.000)
Dwelling with 3 rooms	0.799*** (0.000)	0.798*** (0.000)	0.817*** (0.000)	0.696*** (0.000)	0.766*** (0.000)	0.762*** (0.000)	0.799*** (0.000)	0.798*** (0.000)
Dwelling with 4 rooms	0.984*** (0.000)	0.983*** (0.000)	1.005*** (0.000)	0.871*** (0.000)	0.954*** (0.000)	0.934*** (0.000)	0.985*** (0.000)	0.982*** (0.000)
Dwelling with 5 rooms	1.177*** (0.000)	1.176*** (0.000)	1.195*** (0.000)	1.044*** (0.000)	1.177*** (0.000)	1.140*** (0.000)	1.175*** (0.000)	1.176*** (0.000)
<i>Other control variables:</i>								
Construction Period	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fixed-effects:</i>								
IRIS $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Num.Obs.	973 645	973 645	722 739	132 898	85 845	32 163	366 455	607 190
R2	0.830	0.830	0.822	0.793	0.785	0.803	0.824	0.832
F-Stat		853.59	342.55	209.22	457.69	13.56	727.76	827.01

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: UO stands for urban overheating; standard errors are clustered by RSU; instrument used is the average wind speed; PRS stands for Paris; MSL for Marseille; TLS for Toulouse; MPL for Montpellier; 2013-15 stands for dwellings sold before January the 1st, 2016; 2016-19 for dwellings sold after January the 1st, 2016; Relative density is the density relative to the average density of the IRIS; House dummy is a dummy variable taking 1 if the dwelling a house, 0 otherwise.

7.3 Using an alternative measure of urban overheating

In an optimal situation, we would have precise Land Surface Temperature data available at multiple points in time. Although such data are highly demanded in many scientific fields and by urban governments, technological constraints currently make it difficult to obtain high-resolution Land Surface Temperature data at several temporal points. However, alternative methods exist to measure urban overheating. It is possible to rely on micrometeorological simulations, which provide meteorological indicators at very fine resolution. In the rest of this section, I will present and use data from the UrbanSIS project (Lauwaet et al., 2024).

The UrbanSIS project is designed to study urban overheating in 100 urban areas in Europe using micrometeorological simulations. These simulations are conducted at a very fine resolution of 100 meters by 100 meters. Simulations are based on the UrbClim model, which consists of two modules. The first module is a large-scale atmospheric model that simulates meteorological conditions over the entire urban area. This model is calibrated using data from the European Center for Medium-Range Weather Forecasts. The second module provides a schematic representation of the urban environment at a 100-meter scale. The input data of the urban environment are derived from Corine Land Cover and the Copernicus Land Service. By combining meteorological conditions at the city scale with the physical characteristics of each neighborhood, the authors are able to robustly simulate the air temperature at 2 meters above ground level for every neighborhood of the sampled cities. One of the key features of this dataset is its temporal resolution. Simulations are conducted hourly from January 2007 to December 2021. This is made possible because the atmospheric model is recalibrated every two months using data from the European Center for Medium-Range Weather Forecasts, after which it evolves freely over the next two-months period. The publicly available data include daily averages, minimum, and maximum air temperature at 2 meters above ground, as well as maps of various indicators, obtained by averaging data from 2011 and 2021.

Although these data seem highly useful for my purposes, using them, particularly in an instrumental variable strategy, requires some cautions. These data are generated by meteorological models that incorporate numerous factors at various scales, which raises the risk of tautological reasoning. Specifically, using an instrument in this case risks explaining the simulated temperature variable with a factor that has already been incorporated into the simulations. For this reason, I will rely on two indicators produced by the UrbanSIS team that help reduce this risk: the average annual number of nights where the minimum temperature does not drop below 20°C and the average annual number of days where the maximum temperature exceeds 30°C. These indicators are calculated as averages over the years 2011 to 2020. While they do not vary over time, they remain representative of urban overheating conditions during the period of interest of the housing data. They take the form of rasters with a resolution of 100 meters by 100 meters. Since these two variables are a number of days based on a temperature threshold, they do not directly involve the simulated values from the UrbanSIS project, thereby it reduces the risk of tautological reasoning. Additionally, these variables address concerns related to the fact that Land Surface Temperature (LST) does not directly indicate urban residents' exposure to extreme temperatures. In France and most European countries, nights when the temperature does not drop below 20°C are referred to as tropical nights. Tropical nights are considered extreme climatic events and have significant implications for the comfort and health of both young children and the elderly.⁷ Similarly, days when the maximum temperature exceeds 30°C are referred to as hot days and are associated with numerous effects on humans.⁸ Moreover, because these variables are derived from local temperature conditions, they are influenced by the city's physical characteristics and human activities that together amplify temperatures in urban environments. As such, these variables align well with the definition of urban overheating provided in section 3. Furthermore, although these variables differ significantly from LST, they are highly correlated. The correlation coefficient between the average annual number of nights where the minimum temperature does not drop below 20°C and the LST is 0.758. The correlation coefficient between the average annual number of days where the maximum temperature exceeds 30°C and the LST is 0.926. This indicates that these three measures, although derived from different measurements, generally identify the same neighborhoods in the city as being subject to urban overheating.

Before proceeding, it is important to note that the UrbanSIS project data do not use the same definition of urban areas as the Thermocity data and do not cover exactly the same area as the Thermocity data. This explains the differences in the number of observations compared to the previous analysis. I keep the same identification strategy and the same

⁷<https://www.insee.fr/fr/statistiques/8188144>

⁸<https://meteofrance.com/actualites-et-dossiers/actualites/climat/temperatures-seuils-de-25-degc-et-30-degc-synonymes-de-chaleur>

hedonic price model specification. Tables 9 and 10 present the estimation results for the effect of urban overheating on housing prices using, respectively, the average annual number of nights when the minimum temperature does not drop below 20°C and the average annual number of days when the maximum temperature exceeds 30°C. The estimated coefficient is consistently negative and significant, except for the urban unit of Montpellier. Comparing the results in these tables to those obtained with the LST analysis, we observe that the order of the urban-specific coefficients aligns with the LST analysis. Specifically, the coefficient estimated with 2SLS is higher in absolute value than the coefficient estimated with OLS. Furthermore, the ordering of effects across cities is also consistent. Marseille exhibits the highest predicted effect, Paris the second highest, and Toulouse the lowest. Additionally, the estimated coefficient for the Montpellier urban unit is systematically insignificant. However, with both variables, the coefficients estimated are very similar across periods. The 2SLS coefficients shown in Table 9 are remarkably consistent across samples, around 0.01, suggesting that one additional night per year when the temperature does not drop below 20°C is associated with a 1% decrease in housing prices. Similarly, the 2SLS coefficients in Table 10 are also consistent across samples, around 0.025, implying that one additional day per year when the maximum temperature exceeds 30°C is associated with a 2.5% decrease in housing prices. The results obtained using alternative urban overheating variables appear to confirm those derived from the LST data.

Table 9: OLS and 2SLS estimates of the urban overheating effect on housing prices, with urban overheating measured as the average annual number of nights in which the minimum temperature does not drop below 20°C

	Logarithm of selling price							
	OLS		2SLS					
	FULL (1)	FULL (2)	PRS (3)	MSL (4)	TLS (5)	MPL (6)	2013-15 (7)	2016-19 (8)
NB Night > 20°C	-0.005*** (0.000)	-0.009*** (0.000)	-0.012** (0.024)	-0.013*** (0.000)	-0.008** (0.021)	0.000 (0.965)	-0.010*** (0.000)	-0.009*** (0.000)
Distance to center (km)	0.021*** (0.000)	0.029*** (0.000)	0.025*** (0.005)	0.022 (0.121)	0.039*** (0.000)	0.019 (0.118)	0.031*** (0.000)	0.028*** (0.000)
Relative density	-0.026*** (0.000)	-0.025*** (0.000)	-0.023*** (0.000)	-0.029*** (0.000)	-0.017*** (0.000)	-0.036*** (0.000)	-0.024*** (0.000)	-0.026*** (0.000)
House dummy	0.534*** (0.000)	0.534*** (0.000)	0.524*** (0.000)	0.604*** (0.000)	0.534*** (0.000)	0.447*** (0.000)	0.510*** (0.000)	0.548*** (0.000)
Number of dependencies	0.134*** (0.000)	0.134*** (0.000)	0.139*** (0.000)	0.171*** (0.000)	0.093*** (0.000)	0.064*** (0.000)	0.139*** (0.000)	0.132*** (0.000)
Dwelling with one room	REF	REF	REF	REF	REF	REF	REF	REF
Dwelling with 2 rooms	0.471*** (0.000)	0.471*** (0.000)	0.489*** (0.000)	0.344*** (0.000)	0.415*** (0.000)	0.407*** (0.000)	0.468*** (0.000)	0.473*** (0.000)
Dwelling with 3 rooms	0.815*** (0.000)	0.814*** (0.000)	0.834*** (0.000)	0.671*** (0.000)	0.764*** (0.000)	0.764*** (0.000)	0.814*** (0.000)	0.815*** (0.000)
Dwelling with 4 rooms	1.009*** (0.000)	1.008*** (0.000)	1.031*** (0.000)	0.859*** (0.000)	0.953*** (0.000)	0.935*** (0.000)	1.011*** (0.000)	1.007*** (0.000)
Dwelling with 5 rooms	1.213*** (0.000)	1.212*** (0.000)	1.233*** (0.000)	1.061*** (0.000)	1.184*** (0.000)	1.141*** (0.000)	1.215*** (0.000)	1.211*** (0.000)
<i>Other control variables:</i>								
Construction Period	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fixed-effects:</i>								
IRIS FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Num.Obs.	798 632	798 632	627 703	69 564	69 321	32 044	301 709	496 923
R2	0.829	0.829	0.813	0.781	0.780	0.794	0.827	0.834
F-Stat		1045.56	605.84	306.11	360.14	211.51	835.80	967.68

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: UO stands for urban overheating and is measured with the average annual number of nights in which the minimum temperature does not drop below 20°C; standard errors are clustered by RSU; instrument used is the average wind speed; PRS stands for Paris; MSL for Marseille; TLS for Toulouse; MPL for Montpellier; 2013-15 stands for dwellings sold before January the 1st, 2016; 2016-19 for dwellings sold after January the 1st, 2016; Relative density is the density relative to the average density of the IRIS; House dummy is a dummy variable taking 1 if the dwelling is a house, 0 otherwise.

Table 10: OLS and 2SLS estimates of the urban overheating effect on housing prices, with urban overheating measured as the average annual number of days in which the maximum temperature exceeds 30°C

	Logarithm of selling price							
	OLS		2SLS					
	FULL (1)	FULL (2)	PRS (3)	MSL (4)	TLS (5)	MPL (6)	2013-15 (7)	2016-19 (8)
NB Day > 20°C	-0.010*** (0.000)	-0.025*** (0.000)	-0.025** (0.023)	-0.027*** (0.000)	-0.021** (0.021)	0.004 (0.965)	-0.025*** (0.000)	-0.024*** (0.000)
Distance to center (km)	0.018*** (0.000)	0.024*** (0.000)	0.017*** (0.004)	-0.013 (0.125)	0.037*** (0.000)	0.015 (0.863)	0.026*** (0.000)	0.024*** (0.000)
Relative density	-0.027*** (0.000)	-0.027*** (0.000)	-0.023*** (0.000)	-0.035*** (0.000)	-0.018*** (0.000)	-0.036*** (0.000)	-0.025*** (0.000)	-0.028*** (0.000)
House dummy	0.533*** (0.000)	0.531*** (0.000)	0.523*** (0.000)	0.598*** (0.000)	0.529*** (0.000)	0.449*** (0.000)	0.508*** (0.000)	0.545*** (0.000)
Number of dependencies	0.134*** (0.000)	0.134*** (0.000)	0.139*** (0.000)	0.173*** (0.000)	0.092*** (0.000)	0.064*** (0.000)	0.138*** (0.000)	0.132*** (0.000)
Dwelling with one room	REF	REF	REF	REF	REF	REF	REF	REF
Dwelling with 2 rooms	0.471*** (0.000)	0.471*** (0.000)	0.489*** (0.000)	0.344*** (0.000)	0.413*** (0.000)	0.407*** (0.000)	0.468*** (0.000)	0.473*** (0.000)
Dwelling with 3 rooms	0.815*** (0.000)	0.815*** (0.000)	0.834*** (0.000)	0.672*** (0.000)	0.762*** (0.000)	0.764*** (0.000)	0.815*** (0.000)	0.815*** (0.000)
Dwelling with 4 rooms	1.009*** (0.000)	1.009*** (0.000)	1.031*** (0.000)	0.859*** (0.000)	0.952*** (0.000)	0.935*** (0.000)	1.012*** (0.000)	1.008*** (0.000)
Dwelling with 5 rooms	1.213*** (0.000)	1.213*** (0.000)	1.233*** (0.000)	1.062*** (0.000)	1.183*** (0.000)	1.141*** (0.000)	1.216*** (0.000)	1.212*** (0.000)
<i>Other control variables:</i>								
Construction Period	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fixed-effects:</i>								
IRIS FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Num.Obs.	798 632	798 632	627 703	69 564	69 321	32 044	301 709	496 923
R2	0.829	0.829	0.813	0.781	0.781	0.794	0.827	0.834
F-Stat		488.65	565.70	164.93	166.03	0.98	435.15	468.86

* p < 0.1, ** p < 0.05, *** p < 0.01

Note: UO stands for urban overheating and is measured with the average annual number of days in which the maximum temperature exceeds 30°C; standard errors are clustered by RSU; instrument used is the average wind speed; PRS stands for Paris; MSL for Marseille; TLS for Toulouse; MPL for Montpellier; 2013-15 stands for dwellings sold before January the 1st, 2016; 2016-19 for dwellings sold after January the 1st, 2016; Relative density is the density relative to the average density of the IRIS; House dummy is a dummy variable taking 1 if the dwelling a house, 0 otherwise.

8 Conclusion

This article looks at the impact of urban overheating on housing prices. Urban overheating is a phenomenon that has numerous adverse consequences for urban citizens, is therefore the subject of much research outside the economic field. In this article, I have attempted a first estimation of the effects of urban overheating on housing prices for 4 French urban units, in order to reveal households preferences and to lay the groundwork for a larger discussion on the impact of urban overheating and climate change on urban configurations.

To estimate the effect of urban overheating on housing prices, a hedonic pricing model data is employed, combining the original Thermocity dataset on urban overheating and DV3F data on housing sells for the 2013-2019 period. In order to tackle endogeneity concerns arising from unobservable factors and measurement errors, I adopt an instrument variable method using wind speed averaged at the RSU level as an instrument, expecting that low wind speed will be associated with high land surface temperature. The outcomes of the 2SLS estimation reveal a negative impact of urban overheating on housing prices, suggesting that an increase of 1 degree in Land Surface Temperature is associated with a decrease by 4.2% in housing prices. Examining the sample on two periods indicates a growing effect over time. The city-by-city analysis shows that in Paris and in Toulouse, increasing overheating by 1 degree is associated with a decrease in housing prices by respectively 2.5% and 2.1%. Results are slightly impacted by alternative specification, including location $\times$ Year fixed-effects and land cover variables. Finally, the exploitation of UrbanSIS micrometeorological data tends to confirm the deleterious effect of urban overheating on housing prices.

The results obtained in this article highlight the detrimental effect of urban overheating on the attractiveness of neighborhoods. This finding is particularly important as it provides a strong rationale for urban governments to implement policies aimed at reducing local temperatures, particularly through greening programs and urban renewal initiatives. However, broader reflections are needed to properly integrate these cooling solutions, as such policies may have adverse effects that could undermine their effectiveness. Indeed, greening policies could exacerbate residential segregation by making the most vegetated areas of the city increasingly accessible to wealthier households, while leaving low-income residents might remain concentrated in the hottest urban zones. Further research is needed to better inform local government decision-making.

9 Acknowledgement

I would like to thank the CEREMA for providing access to the DV3F's data relating to property transfers; the Thermocity project teams for making their data freely available. I gratefully acknowledge support from the CNRS/TGIR HUMA-NUM and IN2P3 Computing Center (Lyon - France) for providing computing and data-processing resources needed for this work.

References

- Belgodere, A., & Casamatta, G. (2023). Second home taxation: Effects of the 2015 french reform. *SSRN Electronic Journal*. Retrieved from <https://papers.ssrn.com/abstract=4568947> doi: 10.2139/SSRN.4568947
- Bilal, A., & Rossi-Hansberg, E. (2023). Anticipating climate change across the united states. Retrieved from <https://www.nber.org/papers/w31323> doi: 10.3386/W31323
- Bono, P.-H., Gravel, N., & Trannoy, A. (2008). L'importance de la localisation dans la valorisation des quartiers marseillais. *Economie publique*. Retrieved from <http://journals.openedition.org/economiepublique/6202> doi: 10.4000/ECONOMIEPUBLIQUE.6202
- Bouwknegt, L., & Rouwendal, J. (2023). The effect of urban trees on house prices: evidence from cut-down trees in amsterdam.
- Butsic, V., Hanak, E., & Valletta, R. G. (2011, 2). Climate change and housing prices: Hedonic estimates for ski resorts in western north america. *Land Economics*, 87, 75-91. Retrieved from <https://le.uwpress.org/content/87/1/75> <https://le.uwpress.org/content/87/1/75.abstract> doi: 10.3368/LE.87.1.75
- Chapelle, G., & Eymeoud, J. B. (2022). Can big data increase our knowledge of local rental markets? a dataset on the rental sector in france. *PLoS ONE*, 17. doi: 10.1371/journal.pone.0260405

- Chapelle, G., Eyméoud, J. B., & Wolf, C. (2023). Land-use regulation and housing supply elasticity: evidence from france.
- Cheung, K. Y., Li, R. Y. M., & Shoaib, M. (2018). Walled buildings, sustainability, and housing prices: An artificial neural network approach. *Sustainability* 2018, Vol. 10, Page 1298, 10, 1298. Retrieved from <https://www.mdpi.com/2071-1050/10/4/1298/htm> <https://www.mdpi.com/2071-1050/10/4/1298> doi: 10.3390/SU10041298
- Choi, A., Kim, P., & Park, A. (2023). The effects of air quality on housing prices: Evidence from the aliso canyon gas leak. *Journal of Housing Economics*, 62, 101966. doi: 10.1016/J.JHE.2023.101966
- Coinon, M., Guillain, R., & Legras, S. (2023). We love green: Households' valuation of organic farming and challenges of signaling. Retrieved from <https://papers.ssrn.com/abstract=4542921> doi: 10.2139/SSRN.4542921
- Cvijanovic, D., & de Minne, A. V. (2023, 3). Does climate change affect investment performance? evidence from commercial real estate. *SSRN Electronic Journal*. Retrieved from <https://papers.ssrn.com/abstract=3960805> doi: 10.2139/SSRN.3960805
- Desmet, K., & Rossi-Hansberg, E. (2015). On the spatial economic impact of global warming. *Journal of Urban Economics*, 88, 16-37. doi: 10.1016/J.JUE.2015.04.004
- Durbin, J. (1954). Errors in variables. *Revue de l'Institut International de Statistique / Review of the International Statistical Institute*, 22, 23. doi: 10.2307/1401917
- Exbrayat, N., & Stephane, V. (2024). Does urbanization cause crime? evidence from rural-urban migration in south africa. Retrieved from <https://shs.hal.science/halshs-04390026>
- Frost, C., & Thompson, S. G. (2000). Correcting for regression dilution bias: Comparison of methods for a single predictor variable. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 163, 173-189. Retrieved from <https://dx.doi.org/10.1111/1467-985X.00164> doi: 10.1111/1467-985X.00164
- Grislain-Letrémy, C., Sixou, J., & Sotura, A. (2024). *Urban heat islands and inequalities: Evidence from french cities*.
- Grout, C., Cavailhès, J., Détang-Dessendre, C., & Thomas, A. (2016). Is sprawling residential behavior influenced by climate? *Land Economics*, 92, 203-219. Retrieved from <https://le.uwpress.org/content/92/2/203> doi: 10.3368/LE.92.2.203
- Han, L., Heblich, S., Timmins, C., & Zylberberg, Y. (2024). Cool cities: The value of urban trees. Retrieved from <https://www.nber.org/papers/w32063> doi: 10.3386/W32063
- Hausman, J. A. (2015). Specification tests in econometrics. *Applied Econometrics*, 38, 112-134. doi: 10.2307/1913827
- Hennighausen, H., Liao, Y., Nolte, C., & Pollack, A. (2023). Flood insurance reforms, housing market dynamics, and adaptation to climate risks. *Journal of Housing Economics*, 62, 101953. doi: 10.1016/J.JHE.2023.101953
- Hoehn, J. P., Berger, M. C., & Blomquist, G. C. (1987). A hedonic model of interregional wages, rents, and amenity values. *Journal of Regional Science*, 27, 605-620. Retrieved from <https://onlinelibrary.wiley.com/doi/full/10.1111/j.1467-9787.1987.tb01184.x> doi: 10.1111/J.1467-9787.1987.TB01184.X
- Kang, Y., Niu, D., Song, J., & Sun, W. (2024, 12). Feeling the heat: Extreme temperatures and housing prices in china. *Journal of Economic Surveys*, 38, 1956-1982. Retrieved from <https://onlinelibrary.wiley.com/doi/full/10.1111/joes.12651> <https://onlinelibrary.wiley.com/doi/abs/10.1111/joes.12651> <https://onlinelibrary.wiley.com/doi/10.1111/joes.12651> doi: 10.1111/JOES.12651
- Koirala, B. S., & Bohara, A. K. (2014). Valuing us climate amenities for americans using an hedonic pricing framework. *Journal of Environmental Planning and Management*, 57, 829-847. Retrieved from <https://www.tandfonline.com/doi/abs/10.1080/09640568.2013.770729> doi: 10.1080/09640568.2013.770729
- Lauwaet, D., Berckmans, J., Hooyberghs, H., Wouters, H., Driesen, G., Lefebvre, F., & Ridder, K. D. (2024). High resolution modelling of the urban heat island of 100 european cities. *Urban Climate*, 54. doi: 10.1016/j.uclim.2024.101850
- Li, N., Li, R. Y. M., & Nuttapong, J. (2022). Factors affect the housing prices in china: a systematic review of papers indexed in chinese science citation database. *Property Management*, 40, 780-796. doi: 10.1108/PM-11-2020-0078/FULL/XML
- Luis, J., Olea, M., & Pflueger, C. (2013). A robust test for weak instruments. *Journal of Business Economic Statistics*, 31, 358-369. Retrieved from <https://www.tandfonline.com/doi/abs/10.1080/00401706.2013.806694> doi: 10.1080/00401706.2013.806694

- Luttik, J. (2000). The value of trees, water and open space as reflected by house prices in the netherlands. *Landscape and Urban Planning*, 48, 161-167. doi: 10.1016/S0169-2046(00)00039-6
- Ma, L., & Yildirim, Y. (2023). High temperature, climate change and real estate prices. *SSRN Electronic Journal*. Retrieved from <https://papers.ssrn.com/abstract=4613532> doi: 10.2139/SSRN.4613532
- Maddison, D., & Bigano, A. (2003). The amenity value of the italian climate. *Journal of Environmental Economics and Management*, 45, 319-332. doi: 10.1016/S0095-0696(02)00052-9
- Michel, A., Lonjou, V., Bergsma, E., Leturgie, Q., Roupioz, L., Briottet, X., ... Sabre, M. (2022). Thermocity : generation of an enhanced quality land surface temperature dataset for urban planning. Retrieved from <https://hal.science/hal-04852196>
- Naserikia, M., Hart, M. A., Nazarian, N., Bechtel, B., Lipson, M., & Nice, K. A. (2023). Land surface and air temperature dynamics: The role of urban form and seasonality. *Science of The Total Environment*, 905, 167306. doi: 10.1016/J.SCITOTENV.2023.167306
- Nazarian, N., Krayenhoff, E. S., Bechtel, B., Hondula, D. M., Paolini, R., Vanos, J., ... Santamouris, M. (2022). Integrated assessment of urban overheating impacts on human life. *Earth's Future*, 10, e2022EF002682. Retrieved from <https://onlinelibrary.wiley.com/doi/full/10.1029/2022EF002682> doi: 10.1029/2022EF002682
- Nilsson, P. (2014). Natural amenities in urban space – a geographically weighted regression approach. *Landscape and Urban Planning*, 121, 45-54. doi: 10.1016/J.LANDURBPLAN.2013.08.017
- Pusparum, M., Kurnia, A., & Alamudi, A. (2017). Winsor approach in regression analysis with outlier. *Applied Mathematical Sciences*, 11, 2031. Retrieved from <https://doi.org/10.12988/ams.2017.76214> doi: 10.12988/ams.2017.76214
- Rehdanz, K. (2006). Hedonic pricing of climate change impacts to households in great britain. *Climatic Change*, 74, 413-434. Retrieved from <https://link.springer.com/article/10.1007/s10584-006-3486-5> doi: 10.1007/S10584-006-3486-5/METRICS
- Rehdanz, K., & Maddison, D. (2009). The amenity value of climate to households in germany. *Oxford Economic Papers*, 61, 150-167. Retrieved from <https://dx.doi.org/10.1093/oep/gpn028> doi: 10.1093/OEP/GPN028
- Rizan, A. M., Dennis, L. Y., & Liu, C. (2008). A review on the generation, determination and mitigation of urban heat island. *Journal of Environmental Sciences*, 20, 120-128. doi: 10.1016/S1001-0742(08)60019-4
- Sander, H. A., & Haight, R. G. (2012). Estimating the economic value of cultural ecosystem services in an urbanizing area using hedonic pricing. *Journal of Environmental Management*, 113, 194-205. doi: 10.1016/J.JENVMAN.2012.08.031
- Santamouris, M. (2020). Recent progress on urban overheating and heat island research. integrated assessment of the energy, environmental, vulnerability and health impact. synergies with the global climate change. *Energy and Buildings*, 207, 109482. doi: 10.1016/J.ENBUILD.2019.109482
- Sinha, P., Caulkins, M. L., & Cropper, M. L. (2018). Household location decisions and the value of climate amenities. *Journal of Environmental Economics and Management*, 92, 608-637. doi: 10.1016/J.JEEM.2017.08.005
- Wu, D.-M. (1973). Alternative tests of independence between stochastic regressors and disturbances. *Econometrica*, 41, 733. doi: 10.2307/1914093
- Zhou, D., Xiao, J., Bonafoni, S., Berger, C., Deilami, K., Zhou, Y., ... Sobrino, J. A. (2018). Satellite remote sensing of surface urban heat islands: Progress, challenges, and perspectives. *Remote Sensing 2019, Vol. 11, Page 48, 11*, 48. Retrieved from <https://www.mdpi.com/2072-4292/11/1/48> doi: 10.3390/RS11010048